ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Number Toolkit

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

NUMBER TOOLKIT · CH1 · L1

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The Map

This lesson collapses Number Toolkit into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Prime factor structure

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Standard form conversion

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Fractions and decimals

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Rounding and bounds

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Find the LCM

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 106 website questions, 106 checked solutions, 3 local PDFs read.

01 Prime factor structure

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Number Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$28 = 2^2 \times 7$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q2

Write both numbers as products

$$28 = 2^2 \times 7$$

Find the LCM

$$\text{LCM} = 2^2 \times 3 \times 5 \times 7 = 420$$

Finish: 420

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

2 Find the lowest common multiple (LCM) of 28 and 105

YOUR SOLUTION

02 Standard form conversion

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Number Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

1 Underline the target: what is the question asking you to find?

2 Write the rule, theorem, or equation before any calculator work.

3 Substitute one value at a time and keep units/unknowns visible.

4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^{4} + 7 \times 10^{3} = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^{7}

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> 6 (a) Write 7.8 times 10-4 as an ordinary number. (1) (b) Work out 5 6 10 7 10 2 8 10 4 3 3 . . times + times times - Give your answer in standard form. (2)

YOUR SOLUTION

03 Fractions and decimals

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Number Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3\frac{1}{5} = \frac{16}{5}$$

SIMPLEST STRATEGY

1 Underline the target: what is the question asking you to find?

2 Write the rule, theorem, or equation before any calculator work.

3 Substitute one value at a time and keep units/unknowns visible.

4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q2

Simplify fraction

$$3\frac{1}{5} = \frac{16}{5}$$

Simplify fraction

$$\frac{16}{5} \times \frac{21}{8} = \frac{2 \times 21}{5}$$

Finish: $8\frac{2}{5}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

2 Show that 3 1 5 times 2 5 8 = 8 2 5

YOUR SOLUTION

04 Rounding and bounds

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Number Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> 16 $P = 2a\ c\ d - a$ = 58.4 correct to 3 significant figures. $c = 20$ correct to 2 significant figures. $d = 3.6$ correct to 2 significant figures. Work out the upper bound for the value of P. Show your working clearly. Give your answer correct to 2 decimal places.

YOUR SOLUTION

05 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Number Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

1 Underline the target: what is the question asking you to find?

2 Write the rule, theorem, or equation before any calculator work.

3 Substitute one value at a time and keep units/unknowns visible.

4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^{4} + 7 \times 10^{3} = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^{7}

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> 6 (a) Write 7.8 times 10-4 as an ordinary number. (1) (b) Work out 5 6 10 7 10 2 8 10 4 3 3 . . times + times times - Give your answer in standard form. (2)

YOUR SOLUTION

06 Find the LCM

TRIGGER *"work out", "find", a missing value*

BECOMES A Number Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$28 = 2^2 \times 7$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q2

Write both numbers as products

$$28 = 2^2 \times 7$$

Find the LCM

$$\text{LCM} = 2^2 \times 3 \times 5 \times 7 = 420$$

Finish: 420

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

2 Find the lowest common multiple (LCM) of 28 and 105

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Prime factor structure
"work out", "show", "hence", a familiar exam phrase	Standard form conversion
"work out", "show", "hence", a familiar exam phrase	Fractions and decimals
"work out", "show", "hence", a familiar exam phrase	Rounding and bounds
"work out", "show", "hence", a familiar exam phrase	Convert standard form
"work out", "find", a missing value	Find the LCM

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.